

ARKHANGELSK TALAGI, Russian Federation

WMO: 225500

Lat: 64.5042N

Lon: 40.7278E

Elev: 27

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-19.6	-31.3	0.9	-25.8	-24.7	1.4	-19.5	20.0	21.1	18.1	18.2	2.7	110	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	82.0	67.0	77.4	64.5	73.6	62.4	69.0	78.2	66.7	75.1	64.3	71.2	7.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	95.7	73.1	63.4	87.3	70.6	61.1	80.4	68.4	33.1	78.4	31.3	75.0	29.5	71.1	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.9	15.7	13.9	-31.0	86.9	5.6	3.2	-35.0	89.2	-38.3	91.1	-41.5	92.9	-45.6	95.3		
(5)			-30.9	72.4	5.7	3.0	-35.0	74.5	-38.3	76.3	-41.5	78.0	-45.6	80.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	10.4	11.4	21.0	33.1	44.9	55.0	61.3	56.1	47.5	36.6	24.1	16.8
	DBStd	20.25	15.27	14.97	10.12	8.64	9.63	8.51	7.80	6.79	6.41	7.35	11.29	14.14
	HDD50	6347	1228	1082	899	510	221	41	6	15	121	417	778	1029
	HDD65	11039	1693	1502	1364	956	625	318	167	288	524	880	1228	1494
	CDD50	868	0	0	0	4	63	191	356	204	48	2	0	0
	CDD65	84	0	0	0	0	2	18	52	12	0	0	0	0
	CDH74	797	0	0	0	1	38	200	443	114	1	0	0	0
	CDH80	170	0	0	0	0	6	47	96	21	0	0	0	0
Wind	WSAvg	6.8	6.8	7.0	7.2	7.2	7.6	7.0	6.2	5.9	6.3	6.9	6.9	7.0
Precipitation	PrecAvg	22.8	1.4	1.1	1.1	1.1	1.7	2.3	2.6	2.8	2.2	2.6	1.9	1.7
	PrecMax	28.9	2.6	2.2	2.0	2.8	3.2	4.0	5.4	5.7	4.7	5.0	3.0	2.9
	PrecMin	18.1	0.3	0.3	0.4	0.0	0.6	0.5	0.2	0.7	0.6	0.9	0.9	0.8
	PrecStd	3.2	0.5	0.5	0.4	0.6	0.7	0.9	1.5	1.2	1.0	0.9	0.5	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	37.5	44.5	66.4	78.9	85.7	87.4	82.7	69.5	55.8	44.7	38.9
		MCWB	34.5	35.4	36.8	51.0	58.9	68.3	70.0	68.8	61.1	51.7	43.2	38.0
	2%	DB	33.7	34.5	39.0	55.2	71.7	80.0	82.5	77.0	64.5	52.0	41.3	35.9
		MCWB	33.0	33.4	34.4	46.0	56.3	64.0	67.5	65.6	57.2	49.3	40.5	34.8
	5%	DB	32.0	32.8	36.7	48.9	66.6	75.0	79.0	71.9	60.9	49.0	39.0	34.4
		MCWB	31.4	31.8	33.5	41.3	53.6	61.2	66.3	62.9	55.8	47.3	38.3	33.6
	10%	DB	29.8	30.3	34.6	44.9	62.3	70.3	75.3	68.0	57.6	46.3	36.1	33.4
		MCWB	29.1	29.1	32.4	38.8	52.2	59.0	64.8	61.1	53.7	45.0	35.3	32.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.5	35.7	37.7	52.2	63.1	69.6	73.0	70.8	62.3	53.0	43.5	37.6
		MCDB	34.8	36.8	41.5	63.6	73.5	81.0	81.4	78.6	67.0	55.5	43.9	38.1
	2%	WB	32.8	33.5	35.4	46.3	58.3	66.7	70.0	67.1	59.1	49.9	40.5	34.9
		MCDB	33.2	34.2	37.7	53.7	67.1	77.6	79.3	74.4	63.3	51.5	41.2	35.4
	5%	WB	31.2	31.8	34.0	42.2	55.4	63.3	67.9	64.0	56.6	47.3	38.0	33.7
		MCDB	31.8	32.6	35.8	47.2	65.0	70.6	76.4	69.7	60.0	48.8	38.5	34.3
	10%	WB	28.7	29.0	32.5	39.4	52.3	60.1	65.7	61.7	54.4	44.8	35.5	32.3
		MCDB	29.4	30.0	34.4	43.9	61.0	68.4	73.2	67.4	57.6	45.9	36.0	32.9
Mean Daily Temperature Range	5% DB	MDBR	12.1	13.5	16.1	16.0	17.6	18.2	17.9	16.4	13.4	8.2	8.6	10.5
		MCDBR	9.3	9.8	13.7	21.8	26.2	24.3	23.1	22.0	17.7	10.1	7.7	7.1
		MCWBR	9.1	9.1	10.5	13.6	13.8	11.9	11.6	11.9	11.7	8.4	7.6	6.9
	5% WB	MCDBR	9.1	9.8	10.7	19.8	23.4	21.3	20.8	19.5	15.5	9.7	7.2	7.1
		MCWBR	9.2	9.4	8.6	12.9	13.7	11.5	11.5	11.6	11.1	8.7	7.2	7.2
Clear-Sky Solar Irradiance	taub	0.227	0.287	0.316	0.352	0.357	0.350	0.388	0.364	0.337	0.311	0.356	0.222	
	taud	1.944	2.065	2.107	2.190	2.416	2.472	2.383	2.430	2.491	2.340	2.545	1.953	
	Ebn at Noon	93	187	237	258	270	275	259	254	238	185	90	29	
	Edh at Noon	9	20	30	36	31	30	32	28	22	16	7	3	
All-Sky Solar Radiation	RadAvg	42	204	631	1151	1510	1699	1615	1105	600	219	50	15	
	RadStd	5	38	81	108	140	160	164	126	64	36	10	1	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.13	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page